

Unit ID: 67909623aaf593b2823f0aaa

Unit 3 - Building a Simple Restroom Indicator

PDF Documents

PDF Document 1: 1737528865209-Chapter 5 - Unit 3 - Building a Simple Restroom Indicator.pdf

This PDF document is included in the unit materials.

Update #1 - 2025-05-16 15:17:09



UNIT 3 BUILDING A SIMPLE RESTROOM INDICATOR

In this section, we'll create a simple restroom indicator that helps show if a restroom is occupied or available. We will use a dial to adjust the turn value and LEDs to indicate whether the restroom is in use.

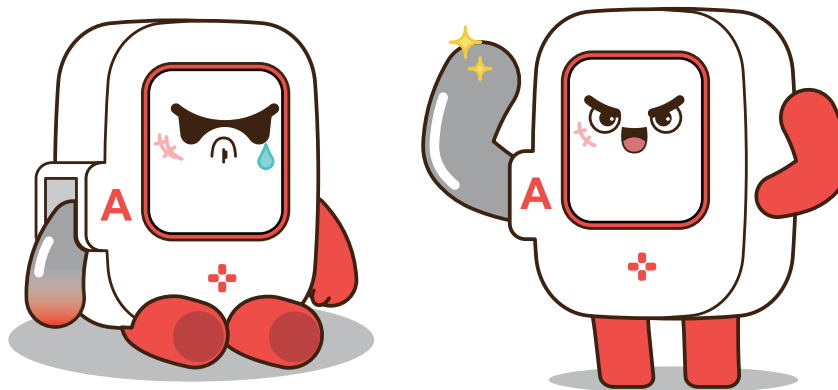
UNDERSTANDING THE RESTROOM INDICATOR

Our restroom indicator uses a dial and two different colored LEDs (red and green) to show the status of the restroom:

DIAL ADJUSTMENT The dial helps us set the turn value, which tells us if the restroom is occupied. The turn value is measured in percentage.

LED LIGHTS We use two LED lights to show the status of the restroom:

- **Red LED** Lights up when the turn value is 50% or more, indicating that the restroom is occupied.
- **Green LED** Lights up when the turn value is 50% or less, indicating that the restroom is available.

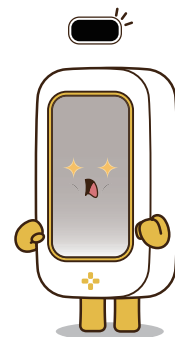
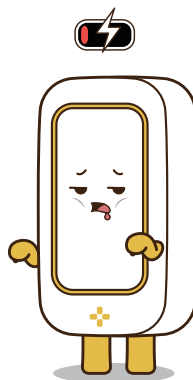
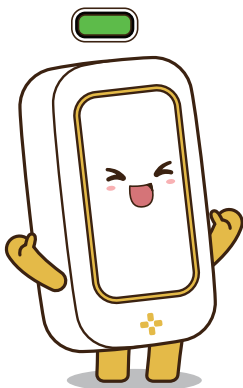


Let's look detail about the **Dial** module.



In Dial, there is 4 values.

1. Turn(%)	2. Angle(°)
You can change the position by turning the knob Range: 0 % ~ 100 %	You can change the angle by turning the knob. Range: 0 ° ~ 360 °





Activity: Create Restroom Indicator

1 Materials Needed:



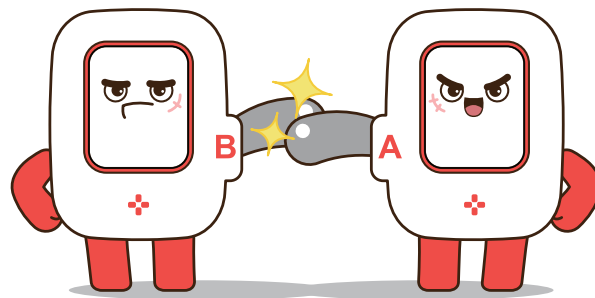
Network



Dial



LED



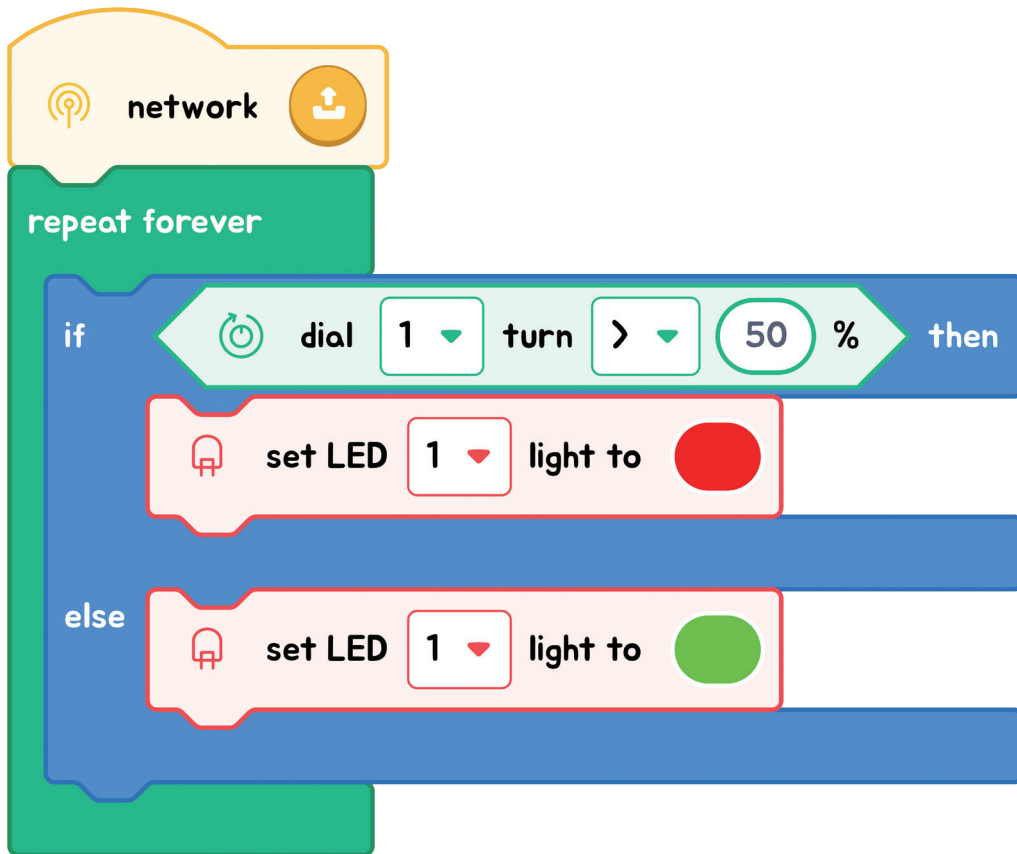
2 Code story

If Dia turn is over than **50%**

↓ true
Set LED as **RED**

↓ false
Set LED as **GREEN**

Code Quest



Checkpoint

If we change the turn what will happen?

50 → 30	50 → 70